



Figure 4: The Properties panel shown on right side of main window

install the Java SDK if you don't have it—get the latest from <http://goo.gl/zXjC>. Instead of downloading these separately, you can also choose to download JDK7u5 + NetBeans from the above mentioned page.

Since this article assumes that readers know core Java and have a working knowledge of NetBeans, its installation and certain other details (yet to come) are left to them.

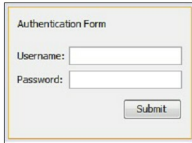


Figure 5: A formatted *Authentication Form*

Implementing two-step verification

Here's a procedure to create a sample authentication form in NetBeans and modify it to implement the two-step verification functionality:

1. Launch NetBeans, select **Files > New Project** or press **Ctrl+Shift+N**.
2. In the **New Project** window, select **Java** under **Categories**, **Java Application** under **Projects** and click **Next**.
3. In the subsequent window, enter the **Project Name** as **Two Step Ver** to the **Project Location**, add **TSV** at the end (this will be the project folder created at the selected location), uncheck the **Create Main Class** check-box and click **Finish**. See Figure 1 for reference. The new project is shown in the **Projects** panel at the left of the main window.
4. Next, create a blank GUI form for authentication. Select **Files > New File** or press **Ctrl+N**. In the **New File** window, select **Swing GUI Forms** under **Categories** and **JFrame Forms** under **File Types** and click **Next**.
5. In the **New JFrame Form** window, enter the **Class Name** as **AuthenticationForm**, the **Package** as **tsv**, and click **Finish**. Observe that **File AuthenticationForm.java** is added to the project under the **tsv** package seen in the **Projects** panel. Also, a blank canvas now appears in the **Workspace** (Figure 2). We'll be using this to place the form elements.
6. On your canvas, add different form elements—four of them (3 **JLabels**, 1 **JTextField**, 1 **JPasswordField** and 1 **JButton**) from **Swing Controls** in the palette on the right

hand side of the main window. Create a basic layout for the **Authentication Form** as shown in Figure 3. Don't worry about the text; you can format it in the next stage. However, you may change the size of the components to make the form look elegant.

7. To format the form, you need to set the text property of each component. Click a component, and in the **Properties** window (at the right of the main window) set the text property. See Figure 4 for an example. Similarly, set the text properties of the remaining components. Leave the text property blank for the **JTextField** and **JPasswordField**. See Figure 5 for reference.
8. Your form is now ready to use, but it's still 'dead'. In order to make it listen, you need to add an event listener for the **Submit** button. When a user clicks it, the listener will first validate the form for empty fields, then extract the text from the **JTextField** and **JPasswordField** and verify it against stored values (for this tutorial, hard-coded in the code itself). If successful, it sends a randomly generated 3-5 digit code to the users' mobile phone and asks them to enter the same code. If the code entered matches the randomly generated code, the users have passed the two-step verification. To add an event, right-click the **Submit** button and select **Events > Action > actionPerformed**. This listener will listen to events triggered via the mouse as well as the keyboard for clicking the **Submit** button. It will create a method for the event listener, where you'll place your event-handling code. It will also switch to **Source** view; to go back to the **Design** view, click **Design**.
9. In the event method, add the following lines of code. This code handles validation, the first step of verification, and it generates a random five-digit code that'll be sent via email-to-SMS to the registered mobile phone number.

```

/*Declaring the container variables*/
String userName, usrPassword;
/*Extract Username and Password from fields*/
userName = jTextField1.getText();
usrPassword = jPasswordField1.getText();
/*Validating against empty fields*/
if(usrName.equals(""))javax.swing.JOptionPane.
showMessageDialog(null, "Please fill the Username Field");
else if(usrPassword.equals(""))javax.swing.JOptionPane.
showMessageDialog(null, "Please fill the Password Field");

/*Implementing first step of verification*/
else {
if(usrName.equalsIgnoreCase("Sample User") && usrPassword.
equals("Samp73")) {
javax.swing.JOptionPane.showMessageDialog(null,"Verified at Step
1");

/*Implementing second step of verification*/
/*Generating random code of 5 digits*/

```